

Assignment questions

UNIT I

1. Derive Schrodinger's time-independent wave equation and explain the significance of the wave function.
2. Derive the expressions for eigenvalues and eigenfunctions for a particle confined in a one-dimensional potential box, and explain how the quantization arises from the boundary conditions.
3. Explain the Kronig-Penny model of solids and show that it leads to energy band structure.
4. Explain the concept of the effective mass of an electron and derive an expression for it.
5. Classify the materials into conductors, semiconductors and insulators based on band theory of solids.

UNIT II

1. Derive an expression for electron concentration in an intrinsic semiconductor.
2. Derive an expression for carrier concentration in p-type semiconductor and explain the variation of Fermi level with temperature and concentration of dopants in p-type semiconductors.
3. Explain the Hall Effect and derive an expression for the Hall coefficient.
4. Derive an expression for hole concentration in an intrinsic semiconductor.
5. Derive an expression for carrier concentration in n-type semiconductor and explain the variation of Fermi level with temperature and concentration of dopants in n-type semiconductors.

UNIT III

1. Explain Dirac's Bra-Ket notation with examples of inner product and outer product.
2. Explain how Bloch's sphere is used to represent the state of a qubit
3. Discuss the role of Hilbert space in quantum computation.
4. Describe how classical computation differs from quantum computation.
5. Explain the geometric interpretation of qubit states on Bloch's sphere.

UNIT IV

1. Explain the classification of magnetic materials.
2. Apply the concept of Weiss domain theory to explain hysteresis behavior in ferromagnetic materials.
3. Explain the differences between soft and hard magnetic materials in terms of hysteresis.
4. Explain the principle of the sol-gel method using a block diagram and its advantages in synthesizing ferrimagnetic materials.
5. Explain different types of dielectric polarization mechanisms in materials.

UNIT V

1. Explain Einstein's coefficients and establish their relations.
2. Describe the construction and working of the ruby laser with necessary diagrams.
3. Explain the construction and working of He-Ne laser with necessary diagrams.
4. Define and derive an expression for acceptance angle for an optical fiber. How it is related to Numerical aperture?
5. Explain working mechanisms of light propagation in step index and graded index optical fibers with suitable diagrams.

Tutorial problems

UNIT-I

1. An electron is localized to $\Delta x = 1.0$ nm. Using $\Delta x \Delta p \geq \hbar/2$, estimate the minimum uncertainty in momentum Δp and the corresponding uncertainty in speed Δv .
2. The de-Broglie wavelength associated with an electron is 0.1 Å. Find the potential difference by which the electron is accelerated.
3. Calculate the energy difference between the ground state and the first excited state for an electron in a box of length 1 Å.

UNIT-II

1. A copper strip carries a current of 3 A in a magnetic field of 0.5 T perpendicular to the strip. The thickness of the strip is 0.1 mm, and the number density of electrons is $8.5 \times 10^{28} \text{ m}^{-3}$. Find the Hall voltage.
2. An intrinsic semiconductor has $E_g = 1.12$ eV, effective mass ratio $m_h^*/m_e^* = 2$, and $T = 300$ K. Calculate $E_c - E_i$ and $E_i - E_v$.
3. For germanium at $T=300$ K, $E_g=0.66$ eV, $N_C=1.0 \times 10^{19} \text{ cm}^{-3}$, $N_V=6.0 \times 10^{18} \text{ cm}^{-3}$. Find n_i .

UNIT-III

1. Express the state $|\psi\rangle = 3|0\rangle + 4|1\rangle$ as a normalized ket.
2. Show that the states $|0\rangle$ and $|1\rangle$ form an orthonormal basis for a single qubit Hilbert space.
3. Represent the qubit state $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ on the Bloch sphere.

UNIT - IV

1. A material has a magnetic flux density (B) of 0.002 T when subjected to a magnetic field strength (H) of 200 A/m. Calculate the permeability (μ) of the material.
2. A paramagnetic material has a relative permeability of 1.0004 . Calculate its magnetic susceptibility (χ).
3. A ferromagnetic material has a magnetic susceptibility of 5000 . Calculate its relative permeability and absolute permeability, given $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$.

UNIT - V

1. The numerical aperture of an optical fiber is 0.39 . If the difference in the refractive indices of the material of its core and the cladding is 0.05 , calculate the refractive index of the material of the core.
2. A step-index fiber has a core refractive index of 1.48 and a cladding refractive index of 1.46 . Calculate the numerical aperture (NA) and acceptance angle.
3. A multimode optical fiber experiences a bending loss of 0.5 dB per loop when bent to a radius of 2 cm. If the fiber is looped 10 times at this radius, what is the total bending loss in dB?

Short Answer Questions

UNIT I

1. Explain de Broglie hypothesis and write an equation for de Broglie wavelength.
2. What is wave particle duality?
3. State Heisenberg's uncertainty principle.
4. Explain the physical significance of wavefunction.
5. State postulates of quantum mechanics.
6. Define an operator in quantum mechanics and give two examples.
7. Define the effective mass of an electron and write an expression for it.
8. Explain the formation of energy bands in solids.
9. State and explain Bloch theorem.

UNIT II

1. Distinguish between intrinsic and extrinsic semiconductors.
2. Distinguish between n-type and p-type semiconductors.
3. Write any four applications of the Hall effect.
4. Explain the effect of temperature on the Fermi level in n-type semiconductors.
5. Explain the effect of temperature on the Fermi level in p-type semiconductors.
6. Explain the Fermi level in an intrinsic semiconductor.
7. Distinguish between direct and indirect band gap semiconductors.

UNIT III

1. What is meant by quantum information processing?
2. Define unitary evolution of quantum systems.
3. What is wavefunction collapse in quantum measurement?
4. Define entanglement with an example of two qubits.
5. Write the matrix for the Hadamard gate.
6. What is the significance of quantum gates in computation?
7. List two challenges in building quantum computers.
8. Mention one advantage of quantum computing over classical computing.
9. What is the role of measurement in quantum algorithms?
10. Write one application of entanglement in quantum technology.

UNIT IV

1. Describe the origin of the magnetic moment.
2. Explain electric polarizability and susceptibility.
3. Mention any three applications of magnetic materials.
4. What is magnetic hyperthermia, and how is it used in cancer treatment?
5. What are ferroelectric, piezoelectric, and pyroelectric materials?
6. What is Ferroelectric Random-Access Memory (Fe-RAM)?

UNIT V

1. Explain spontaneous and stimulated emission processes.
2. Explain population inversion in lasers.
3. What are the characteristics of a laser?
4. Write any four applications of lasers.
5. Explain the basic principle used in optical fibers for the transmission of light.
6. Write any four major advantages of optical fiber communication over other conventional communication systems.

Long Answer Questions

UNIT I

1. Explain de Broglie wavelength and derive an equation for de Broglie wavelength in terms of kinetic energy.
2. Derive Schrodinger's time-independent wave equation and explain the significance of the wave function.
3. Derive eigenvalues and eigenfunctions for a particle in a one-dimensional deep potential box.
4. Explain the Kronig-Penney model of solids and show that it leads to energy band structure.
5. Explain the concept of the effective mass of an electron and derive an expression for it.

6. Classify the materials into conductors, semiconductors and insulators based on band theory of solids.

UNIT II

1. Derive an expression for carrier concentration in n-type semiconductor and explain the variation of Fermi level with temperature and concentration of dopants in n-type semiconductors.
2. Derive an expression for carrier concentration in p-type semiconductor and explain the variation of Fermi level with temperature and concentration of dopants in p-type semiconductors.
3. State and explain the Hall effect. Apply the concept of the Hall effect to derive the expression for the Hall coefficient of a given material.
4. Derive an expression for electron concentration in an intrinsic semiconductor.
5. Derive an expression for hole concentration in an intrinsic semiconductor.

UNIT III

1. Discuss how entanglement arises in multi-qubit systems using Bra–Ket notation
2. Explain the role of vector spaces in representing quantum states with examples
3. Explain how quantum computing systems are used for information processing and how they differ from classical systems.
4. Discuss the evolution of quantum systems under unitary transformations with examples.
5. Describe the different types of quantum gates with suitable matrix representations.
6. Compare the efficiency of quantum information processing with classical information processing.
7. Explain the challenges faced in the practical implementation of quantum computing.
8. Discuss the advantages of quantum computing over classical computing with real-world applications.
9. Explain how entanglement is used in quantum communication and quantum teleportation.
10. Describe how quantum gates operate on single and multiple qubits with examples.

UNIT IV

1. Explain the classification of magnetic materials.
2. Explain the Ferromagnetic hysteresis loop on the basis of the domain theory of ferromagnetism.
3. Differentiate soft and hard magnetic materials and mention any two applications of each.
4. Explain the principle of the sol–gel method using a block diagram and its advantages in synthesizing ferrimagnetic materials.
5. Explain different types of dielectric polarization mechanisms in materials.
6. Derive an expression for the Bohr magneton.

UNIT V

1. Describe the construction and working of the ruby laser with necessary diagrams.
2. Explain the construction and working of the He-Ne laser with necessary diagrams.
3. Explain the construction and working of a CO₂ laser with necessary diagrams
4. Explain the acceptance angle and derive an expression for the acceptance angle of an optical fiber. How is it related to Numerical aperture?
5. Differentiate between step index and graded index optical fibers
6. Explain Einstein's coefficients and establish their relations.

Multiple Choice Questions

UNIT I

1. The de Broglie wavelength formula is
 - a. $\lambda = h/p$
 - b. $\lambda = hp$
 - c. $\lambda = p/h$
 - d. $\lambda = h^2p$
2. According to de Broglie, matter waves are more significant for
 - a. Large objects moving slowly
 - b. Microscopic particles moving slowly
 - c. **Microscopic particles moving fast**
 - d. Large objects moving fast
3. Equation for the normalized wave function of a particle in a 3D potential box of length L is _____
 - a. $\int_0^L \int_0^L \int_0^L |\psi|^2 dx dy dz = 1$
 - b. $\int_0^L \int_0^L \int_0^L |\psi \psi^*|^2 dx dy dz = 0$
 - c. $\int_0^L \int_0^L \int_0^L \psi = 0$
 - d. $\int_0^L \int_0^L \int_0^L \psi = 1$
4. $E =$ _____ for a particle in 1D potential box .
 - a. $\frac{n^2 h^2}{8mL^2}$
 - b. $\frac{8mL^2}{n^2 h^2}$
 - c. $\frac{n^2 h^2}{8m^2 L^2}$
 - d. $\frac{n^2 h^2}{8mL}$
5. If a wave function is not normalized, the expectation value formula:
 - a. Still works without changes
 - b. **Must be divided by $\int |\psi|^2 dx$**
 - c. Becomes meaningless
 - d. Gives zero
6. For a normalized wave function, the total probability over all space is:
 - a. 0
 - b. **1**
 - c. Infinity
 - d. Depends on the particle mass
7. In quantum mechanics, the expectation value is interpreted as:
 - a. The most probable value
 - b. **The average result over many measurements**
 - c. The classical value of the variable
 - d. The exact value in one measurement

8. The expectation value of momentum $\langle p \rangle$ in position space is:
- $\int \psi^* (-i\hbar \partial/\partial x) \psi dx$
 - $\int \psi^* i\hbar \partial/\partial x \psi dx$
 - $\int p |\psi|^2 dx$
 - $m \langle v \rangle$
9. According to the Bloch theorem, the potential variation on the electron in the lattice is
- Periodic**
 - Non-periodic
 - Constant
 - Uneven
10. The discontinuities in E-k curve, occurs at
- $K = \pm \frac{n\pi}{a}$
 - $K = \pm \frac{n^2\pi}{a}$
 - $K = \pm \frac{n\pi^2}{a}$
 - $K = \pm \frac{n\pi}{a^2}$
11. Quantum confinement effects become significant when the particle size is:
- Much larger than the Bohr exciton radius
 - Comparable to or smaller than the Bohr exciton radius**
 - Equal to 1 cm
 - Infinite
12. A semiconductor has _____ temperature coefficient of resistance.
- Negative**
 - Positive
 - Zero
 - None of the above
13. According to the Kronig-Penney model, the width of the allowed energy band increase with increasing _____.
- αa**
 - βb
 - Scattering potential
 - None of the above

UNIT II

1. A semiconductor generally has _____ valence electrons
- 2
 - 3
 - 4**
 - 6

2. With an increase in temperature, the resistivity of a semiconductor
 - a. Remains constant
 - b. Increases
 - c. Decreases**
 - d. Unevenly varies
3. A semiconductor is formed by _____ bonds
 - a. Covalent**
 - b. Electrovalent
 - c. Co-ordinate
 - d. None of the above
4. A semiconductor has temperature coefficient of resistance.
 - a. Positive
 - b. Zero
 - c. Negative**
 - d. None of the above
5. The most commonly used semiconductor is
 - a. Germanium
 - b. Silicon**
 - c. Carbon
 - d. Sulphur
6. When a pentavalent impurity is added to a pure semiconductor, it becomes
 - a. An insulator
 - b. An intrinsic semiconductor
 - c. p-type semiconductor
 - d. n-type semiconductor**
7. In a pure semiconductor, the electrical conductivity is governed by the motion of _____.
 - a. Electrons and hole**
 - b. Only holes
 - c. Only electrons
 - d. None of the above
8. If metal or semiconductor carrying current is placed in a magnetic field perpendicular to the current, an electric field is induced in the direction perpendicular to both current and magnetic field. This phenomenon is known as
 - a. Chemical Effect
 - b. Toxicity
 - c. Hall effect**
 - d. None
9. The highest energy level of the filled energy state at 0K is called _____.
 - a. Fermi energy**
 - b. valence band
 - c. conduction band
 - d. Donor energy
10. To get p-type semiconductor _____ impurities are added to Si.
 - a. Trivalent**
 - b. Penta valent
 - c. fourth group elements
 - d. None of the above

11. In n -type material _____ are called the majority carrier.
 - a. holes
 - b. Electrons**
 - c. bosons
 - d. None of the above
12. In an intrinsic semiconductor always _____ number of electrons and holes will present.
 - a. Equal**
 - b. More
 - c. Less
 - d. None of the above
13. _____ is an example for an intrinsic semiconductor.
 - a. Silicon**
 - b. Barium
 - c. Selenium
 - d. Gallium
14. The electron vacancies in a valance band of a semiconductor will exist as _____.
 - a. Hole**
 - b. Boson
 - c. Neutron
 - d. None of the above
15. In p -type material _____ are called the majority carrier
 - a. Electrons
 - b. holes**
 - c. Bosons
 - d. None of the above

UNIT-III

1. In Bra–Ket notation, $\langle \psi | \psi \rangle$ is equal to:
 - a. 0
 - b. 1 (if normalized)**
 - c. $|\psi|^2$
 - d. Complex number
2. The state $|0\rangle + |1\rangle$ normalized is represented as:
 - a. $|0\rangle + |1\rangle$
 - b. $(1/\sqrt{2})(|0\rangle + |1\rangle)$**
 - c. $|0\rangle - |1\rangle$
 - d. $|1\rangle$
3. Hilbert space for n qubits has dimension:
 - a. n
 - b. $2n$
 - c. 2^n**
 - d. n^2
4. The Bloch sphere represents:
 - a. A classical bit
 - b. A qubit state**
 - c. An entangled system
 - d. Tensor product space

5. Classical bits can take values:
 - a. 0 or 1**
 - b. Any real number
 - c. Complex values
 - d. 0, 1, or superpositions
6. Qubits can exist in:
 - a. Only $|0\rangle$
 - b. Only $|1\rangle$
 - c. Any superposition of $|0\rangle$ and $|1\rangle$**
 - d. Only entangled states
7. The poles of Bloch sphere correspond to:
 - a. $|0\rangle$ and $|1\rangle$**
 - b. Superposition states
 - c. Classical bits
 - d. Tensor products
8. A two-qubit system is represented by:
 - a. $|0\rangle, |1\rangle$
 - b. $|00\rangle, |01\rangle, |10\rangle, |11\rangle$**
 - c. $|0\rangle + |1\rangle$
 - d. Bloch sphere
9. Bra–Ket notation was introduced by:
 - a. Schrödinger
 - b. Dirac**
 - c. Einstein
 - d. Heisenberg
10. The inner product of two orthogonal states is:
 - a. 1
 - b. 0**
 - c. ∞
 - d. Undefined
11. Quantum information is processed using:
 - a. Bits
 - b. Qubits**
 - c. Classical registers
 - d. Transistors
12. The evolution of quantum systems is governed by:
 - a. Random noise
 - b. Unitary operators**
 - c. Classical logic
 - d. Probability laws
13. Measurement of a qubit causes:
 - a. Superposition to remain
 - b. Collapse of the wavefunction**
 - c. Entanglement to increase
 - d. No change

14. Entanglement is a property of:
 - a. Independent qubits
 - b. Correlated quantum states**
 - c. Classical states
 - d. Unitary gates
15. One of the major challenges in quantum computing is:
 - a. High transistor density
 - b. Decoherence**
 - c. Heat dissipation
 - d. Binary representation
16. Quantum computing has an advantage over classical computing in solving:
 - a. Addition problems
 - b. Multiplication problems
 - c. Prime factorization**
 - d. Basic logic operations
17. Which gate creates superposition?
 - a. Z gate
 - b. X gate
 - c. H gate**
 - d. CNOT gate
18. Entanglement is essential for:
 - a. Classical computation
 - b. Quantum teleportation**
 - c. Binary arithmetic
 - d. Logic gate design

UNIT IV

1. Dielectrics are
 - a. Solids
 - b. Metals
 - c. Semiconductors
 - d. Insulators**
2. The property of becoming electrically polarized when mechanical stress is applied is known as
 - a. Ferro-electric
 - b. Piezoelectric**
 - c. Pyro-electric
 - d. None of these
3. Orientation polarization is due to _____ of polar molecules in dielectric substances
 - a. Rotation**
 - b. Creation of dipoles
 - c. Charge separation
 - d. None of these

4. The material in which spontaneous polarization changes with temperature is known as
 - a. Piezo-electric
 - b. Ferroelectric
 - c. Pyroelectric**
 - d. Magneto optic
5. The polarizations that are dependent on temperature are
 - a. Orientation and space charge**
 - b. Electronic and ionic
 - c. Orientation and ionic
 - d. Electronic and space charge
6. The transition from ferromagnetic to the paramagnetic state is named after
 - a. Curie**
 - b. Curie-Weiss
 - c. Neel
 - d. Debye
7. Hard magnetic materials have _____ hysteresis loss
 - a. Large**
 - b. Small
 - c. Zero
 - d. None of the above
8. One advantage of the sol-gel method for ferrimagnetic materials is:
 - a. Low cost and room-temperature processing**
 - b. High energy consumption
 - c. Requires high vacuum
 - d. Produces large grain sizes only
9. Which step in sol-gel synthesis controls particle size and homogeneity?
 - a. Drying
 - b. Gelation
 - c. Hydrolysis and condensation**
 - b. Calcination
10. The crystal structure of most ferrites is:
 - a. Hexagonal
 - b. Spinel**
 - c. Cubic close-packed
 - d. Orthorhombic
11. Ferrimagnetic materials are widely used in:
 - a. Superconductors
 - b. Permanent magnets, transformers, and microwave devices**
 - c. Optical fibers
 - d. Photovoltaics

12. Which parameter affects the magnetic properties most in sol-gel ferrites?
- Drying temperature
 - pH of solution, calcination temperature, and particle size**
 - Shape of beaker
 - Stirring speed only
13. The local electric field is calculated using the method suggested by
- Lorentz**
 - Weiss
 - Curie
 - Coulomb
14. Which among the following is a dielectric?
- Copper
 - Aluminum
 - Iron
 - Ceramic**
15. Fe-RAM stores data using which property of materials?
- Electrical conductivity
 - Ferroelectric polarization**
 - Magnetic flux density
 - Photoconductivity
16. Which material is commonly used in Fe-RAM capacitors?
- Silicon dioxide
 - Lead zirconate titanate (PZT)**
 - Aluminum oxide
 - Gallium arsenide
17. A load cell is primarily used to measure:
- Voltage
 - Temperature
 - Force/Weight**
 - Pressure
18. A fire sensor is primarily used to detect:
- Gas flow
 - Smoke, heat, or flame**
 - Sound waves
 - Electrical resistance

UNIT V

1. Emission of a photon when an electron jumps from a higher energy state to a lower energy state due to the interaction of external energy is called
- Spontaneous emission
 - Stimulated emission**
 - Amplified emission
 - None

2. A three-level laser system will be
 - a. Always CW
 - b. Either CW or pulsed
 - c. Always pulsed**
 - d. Made CW as well as pulsed by temperature control
3. The wavelength of emission of CO₂ laser is
 - a. 632.8nm
 - b. 10.6μm**
 - c. 1.064 μm
 - d. 694.3nm
5. The wavelength of emission of the Ruby laser is
 - a. 632.8nm
 - b. 10.6μm
 - c. 1.064 μm
 - d. 694.3nm**
6. The ratio of the mixture of CO₂: N₂: He used in CO₂ laser is
 - a. 5:4:1
 - b. 4:5:1
 - c. 1:4:5**
 - d. 1:5:4
7. The colour of the laser output from a Ruby laser is
 - a. Green
 - b. Blue
 - c. Red**
 - d. violet
8. In He- Ne laser, the ratio of He:Ne is
 - a. 1:10
 - b. 1:1
 - c. 100:1
 - d. 10:1**
9. An example of the creation of population inversion by optical pumping is
 - a. He-Ne laser
 - b. Diode laser
 - c. Ruby laser**
 - d. CO₂ laser
10. Propagation of light through fiber core is due to
 - a. Diffraction
 - b. Interference
 - c. Total internal reflection**
 - d. refraction
11. Choose the correct statement
 - a. Cladding has higher refractive index than core
 - b. Core has higher refractive index than cladding**
 - c. Cladding is for providing mechanical strength to the fiber
 - d. The loss in waveguides without cladding is very low

12. Total internal reflection takes place when light ray travels from
 - a. **Denser to rarer medium**
 - b. Denser to denser medium
 - c. Rarer to rarer medium
 - d. Rarer to denser medium
13. In a fiber, light travels inside
 - a. **Core medium**
 - b. Cladding medium
 - c. Buffer medium
 - d. Air medium
14. The full form of LiDAR is:
 - a. **Light Detection and Ranging**
 - b. Laser Data and Recognition
 - c. Linear Distance Analysis Radar
 - d. Light Diffusion and Reflection
15. In autonomous vehicles, LiDAR is mainly used for:
 - a. Entertainment systems
 - b. **Obstacle detection and environment mapping**
 - c. Engine monitoring
 - d. Weather prediction

Fill in the blanks

UNIT I

1. Dual nature of matter waves was proposed by _____
2. For a normalized wave function, the total probability over all space is _____
3. In Dirac notation, the expectation value of an operator $\hat{A}$ is _____
4. de Broglie wave length $\lambda =$ _____
5. When a potential difference of 400 volts is applied in accelerating an electron, the wavelength attained by an electron is _____
6. One dimensional Schrodinger time independent wave equation is _____
7. According to _____ principle, it is not possible to know position and momentum of a particle at same time.
8. If ψ is an eigenfunction of operator $\hat{A}$ with eigenvalue a , then $\langle A \rangle$ is _____
9. In a particle-in-a-box model, the energy levels are proportional to _____.
10. The Schrödinger wave equation is used to determine _____ of a particle.
11. Δx , Δp are the uncertainties of its position and momentum respectively, then these can be related as _____
12. First Brillouin zone corresponds to k value extending from _____ to _____.
13. Expression of electron effective mass $m^* =$ _____.

Key

- | | | |
|--------------------------------------|---|---|
| 1. de Broglie | 6. $\frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} (E - V) \psi = 0$ | 11. $\Delta x \cdot \Delta p \geq \hbar/4\pi$ |
| 2. 1 | 7. Heisenberg uncertainty | 12. $-\pi/a$ to π/a |
| 3. $\langle \psi A \psi \rangle$ | 8. Equal to a | 13. $\hbar^2 / \frac{d^2 E}{dk^2}$ |
| 4. $\hbar/p = \hbar/mv$ | 9. n^2 | |
| 5. $0.61 \text{ \AA}$ | 10. Probability distribution | |

UNIT II

1. At absolute 0 K, a semiconductor behaves as _____.
2. With increase in temperature, the resistivity of a semiconductor _____.
3. In a pure semiconductor, the electrical conductivity is governed by the motion of _____.
4. In comparison with an intrinsic semiconductor, the resistivity of an extrinsic semiconductor is _____.
5. As the temperature of semiconductor increases, the probability to find an electron in the conduction band _____.
6. The electron vacancies in a valance band of a semiconductor will exist as _____.
7. Expression for Hall coefficient is equal to _____.
8. To get p-type semiconductor _____ elements are added to Si.
9. The highest energy level of the filled energy state at 0K is called _____.
10. _____ is an example for intrinsic semiconductors.
11. The effect which helps us in identifying the type of semiconductor is _____.
12. In an intrinsic semiconductor always _____ number of electrons and holes will present.

Key

- | | | |
|-----------------------------|-----------------------|-----------------|
| 1. Insulator | 5. Increases | 10. Ge, Si |
| 2. Decreases | 6. Hole | 11. Hall effect |
| 3. Both electrons and Holes | 7. $R=1/ne$ | 12. Equal |
| 4. Less | 8. Penta valent | |
| | 9. Fermi energy level | |

UNIT III

1. In Bra–Ket notation, $|\psi\rangle$ is called a _____.
2. The dual of a ket $|\psi\rangle$ is written as _____.
3. The inner product $\langle \psi | \psi \rangle$ equals _____ for normalized states.
4. The Hilbert space dimension of 3 qubits is _____.
5. Bloch sphere is a geometrical representation of a _____.
6. A classical bit can take the values _____.
7. A qubit can exist in a _____ of $|0\rangle$ and $|1\rangle$.
8. The poles of Bloch sphere correspond to states _____ and _____.

9. A two-qubit system is represented in a _____ dimensional space.
10. Dirac's notation is used to simplify representation of _____ states.
11. Quantum information is stored in units called _____.
12. _____ operators describe the evolution of quantum states.
13. Measurement causes _____ of the wavefunction.
14. Entanglement is a unique feature of _____ systems.
15. The Pauli-X gate acts as a _____ operation on qubits.
16. The Hadamard gate creates _____ states from basis states.
17. A CNOT gate acts on _____ qubits.
18. The biggest challenge in quantum computing is _____.
19. One advantage of quantum computing is faster _____ factorization.
20. Quantum teleportation relies on the principle of _____.

Key

- | | | |
|-----------------------|--------------------------------|-------------------|
| 1. Ket | 8. $ 0\rangle$ and $ 1\rangle$ | 15. NOT |
| 2. Bra $\langle\psi $ | 9. 4 | 16. Superposition |
| 3. 1 | 10. Quantum | 17. Two |
| 4. 8 (2^3) | 11. Qubits | 18. Decoherence |
| 5. Qubit | 12. Unitary | 19. Prime |
| 6. 0 or 1 | 13. Collapse | 20. Entanglement |
| 7. Superposition | 14. Quantum | |

UNIT IV

1. The process of producing electric dipoles which are oriented along the field direction is called _____ in dielectrics.
2. The displacement of cations and anions in opposite direction in ionic solids result in _____ polarization.
3. In some materials which have molecules with permanent dipole moment, applied electric field results in orienting these molecular dipoles along the field direction. This phenomenon is called _____ polarization.
4. Electronic polarizability $\alpha_e =$ _____
5. Ionic polarizability $\alpha_i =$ _____
6. In some materials ions diffuse towards the electrodes in response to the applied field resulting in the redistribution of charges in the medium. Such phenomenon is called _____ polarization.
7. The unit of magnetic induction is _____
8. The unit of Magnetic field intensity is _____
9. When the magnetic material is placed inside a magnetic field, if the magnetic lines of force are repelled then the material is _____
10. Area enclosed by the hysteresis loop represents _____.
11. The value of Bohr magneton is equal to _____
12. The lag of magnetization behind the magnetizing field in ferromagnetic materials is known as _____.

Key

- | | | |
|--|--------------------------------|------------------------------------|
| 1. Polarization | | 11. $\mu_B = 9.27 \times 10^{-24}$ |
| 2. Ionic | 6. Space Charge | ampere-m ² |
| 3. Orientational | 7. Tesla, Weber/m ² | 12. Magnetic hysteresis |
| 4. $4\pi\epsilon_0 R^3$ | 8. A/m ² | loop |
| 5. $\alpha_i = \frac{\mu}{E} = \frac{e^2}{w_0^2} \left[\frac{1}{M} + \frac{1}{m} \right]$ | 9. Diamagnetic material | |
| | 10. Hysteresis loss | |

UNIT V

1. LASER is an acronym for Light Amplification by _____ Emission of Radiation
2. When the population of the excited state is larger than the population of the lower state, the condition is called _____.
3. Ruby laser is the example for a _____ level system
4. He-Ne laser is a good example for a _____ level system
5. Wavelength of emission of semiconductor diode laser is _____
6. Wavelength emission of He-Ne laser is _____
7. In He-Ne laser, the active centers are _____
8. Laser radiation is highly _____.
9. Fiber optic communication is based on the phenomenon of _____
10. For total internal reflection to occur, light must travel from denser to rarer medium and the angle of incidence must be greater than _____
11. Light launched at the fiber end within _____ alone propagates to the other end
12. Sine of the maximum acceptance angle is called _____ of the fiber
13. In optical fibers attenuation is mainly due to two sources of losses namely _____ and _____
14. Fractional index changes $\Delta =$ _____

Key

- | | | |
|-------------------------|---------------------|------------------------|
| 1. Stimulated | 7. Ne Atoms | 12. Numerical Aperture |
| 2. Population Inversion | 8. Coherent | 13. Scattering and |
| 3. Three | 9. Total Internal | Bending Losses |
| 4. Four | Reflection | 14. $n_1 - n_2/n_1$ |
| 5. 840 nm | 10. Critical Angle | |
| 6. 632.8 nm | 11. Acceptance Cone | |

Match the following

UNIT I

- | | |
|----------------------------|--|
| 1. Position operator | a) $-i\hbar \frac{\partial}{\partial x}$ |
| 2. Momentum operator | b) $-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}$ |
| 3. Kinetic Energy operator | c) x |

Key:

1 → c 2 → a 3 → b

UNIT II

- | | |
|---|----------------------------|
| 1. Fermi level in P-type semiconductors. | a. Near to conduction band |
| 2. Fermi level in N-type semiconductors. | b. Near to conduction band |
| 3. Fermi level in intrinsic semiconductors. | c. Near to conduction band |

Key:

1 → c 2 → a 3 → b

UNIT-III

- | | |
|------------------|---|
| 1. Hadamard Gate | a. Flips the state of a qubit (NOT operation) |
| 2. Pauli-X Gate | b. Creates a superposition from basis states |
| 3. CNOT Gate | c. Conditional operation on two qubits |

Key:

1 → b 2 → a 3 → c

UNIT-IV

- | | |
|--------------------------|--|
| 1. Magnetic hyperthermia | a. Ferroelectric memory device |
| 2. Fe-RAM | b. Cancer treatment using magnetic nanoparticles |
| 3. Load cell. | c. Sensor for measuring force |

Key:

1 → b 2 → a 3 → c

UNIT-V

- | | |
|-------------------------|---|
| 1. Ruby Laser | a. Used in barcode scanners |
| 2. LIDAR | b. Principle of total internal reflection |
| 3. Optical Fibre Sensor | c. Structural health monitoring |

Key:

1 → a 2 → b 3 → c